

NLP - Project Report

Analysis of Media Narratives in the Reporting of US Actions Against Nicolás Maduro

Alizée Archereau

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Introduction

Context

In January 2026, US forces carried out a military operation in Caracas with the objective of capturing and extraditing Venezuelan President Nicolás Maduro to the United States. The operation was justified by US authorities on the basis of long-standing charges related to narco-terrorism and drug trafficking. Given Maduro's political status and the history of strained relations between the United States and Venezuela, the event immediately drew widespread international media attention.

Coverage of the operation extended beyond factual reporting and quickly became embedded in broader political narratives. As a result, the event offers a useful case study for examining how media outlets with different ideological orientations frame the same geopolitical action.

Research Question (RQ)

The central research question of this project is the following:

How do left-wing and right-wing English-language news media differ in sentiment and topic framing when reporting on US actions against Nicolás Maduro?

Rather than assuming ideological bias a priori, the objective is to assess whether measurable linguistic and thematic differences emerge when both sides report on the same event.

To address this question, the analysis is structured around three complementary sub-questions:

1. **Sentiment**

Is there a significant difference in emotional tone between left-leaning and right-leaning coverage, or are differences primarily driven by vocabulary choice rather than overall sentiment?

2. **Topic Modeling**

Do the two groups emphasize different themes when describing the event, such as military action, legal justification, or diplomatic and economic consequences?

3. **Named Entity Recognition (NER)**

Which actors are most prominently mentioned by each side, and how do these choices shape the narrative by highlighting different political or institutional "protagonists"?

Together, these three perspectives allow for a more nuanced analysis of media framing, combining emotional tone, thematic focus, and narrative structure.

Part 0: Data Collection

Source Selection

The dataset is composed of articles published by 18 major English-language news outlets. These outlets were categorized by ideological orientation using widely referenced media bias charts such as AllSides and Ad Fontes. This classification is not meant to be exhaustive or definitive, but it provides a practical framework for comparing left-leaning and right-leaning coverage of the same event.

- Left-leaning outlets include sources such as **CNN**, **The New York Times**, **Al Jazeera**, and **The Guardian**.
- Right-leaning outlets include **Breitbart**, **Fox News**, the **New York Post**, and **The Daily Wire**.

This selection combines mainstream media with more explicitly partisan outlets, allowing the analysis to capture differences in editorial tone and framing across ideological groups.

Methodology

1. **Article collection** was carried out using a custom Python scraping pipeline.
2. **URLs** were retrieved using the *duckduckgo_search* library, which was chosen to avoid personalization effects commonly used by other search engines. Using DuckDuckGo ensures that the same search queries return comparable results across runs and outlets.
3. Once URLs were collected, articles were **downloaded and parsed using the *newspaper3k*** library. This step extracts the main article text while removing most boilerplate content such as menus and advertisements.

In total, 188 full-text articles were scraped and retained for analysis. Only publicly accessible content was used, and no paywalls were bypassed during data collection. The scraping process was limited to published news articles and basic metadata, in line with usual ethical guidelines for academic research.

Part 1: Data Hygiene & Pipeline Setup

Getting rid of the Garbage

Before trying to answer questions about sentiment or framing, we first need to ensure that the dataset is usable. As is often the case with web-scraped data, the raw articles contain **various forms of noise**. Even when scraping from professional news websites, the extracted text may include encoding issues, formatting artifacts, or fragments that are not part of the actual article content.

The goal of this section is therefore not to “over-process” the data, but to make sure that obvious problems are fixed while preserving as much linguistic structure as possible for later analyses.

1. **Standardize**

The first step consists in fixing basic encoding issues introduced during scraping. Some articles contain improperly decoded characters, for instance accented characters such as “*NicolÃjs*” instead of “*Nicolás*”. These issues were corrected to ensure textual consistency and avoid splitting identical entities into different tokens.

2. **Clean**

We then removed clear scraping artifacts that do not carry semantic meaning. This includes leftover URLs, excessive whitespace, line breaks, and non-informative symbols. At this stage, the cleaning remains conservative and does not alter sentence structure or word order.

3. **Filter**

Duplicate articles were removed to avoid counting the same content multiple times, especially in the case of syndicated news. In addition, very short articles (under 50 words) were excluded from the dataset, as they generally lack sufficient context for sentiment, framing, or topic-based analysis.

4. **Preserve**

Importantly, we deliberately avoid heavy preprocessing at this stage. Stopwords are not removed, and text is not lowercased. As discussed in class, preprocessing choices are task-specific: advanced transformer-based models and Named Entity Recognition rely on capitalization and sentence structure to function correctly. Preserving the original text therefore allows us to reuse the same dataset across multiple NLP tasks without degrading model performance.

Part 2: Lexical Analysis & The "Framing" Check

Methodology: Bigrams instead of unigrams

A simple frequency analysis based on single words is often **insufficient** to capture media framing. Many unigrams are ambiguous and can be used across ideological lines without carrying a clear evaluative stance. Words such as “*capture*”, “*operation*”, or “*extradition*” may describe factual elements of the event, but they do not reveal how the event is framed.

To address this limitation, the analysis focuses on **bigrams** and, when relevant, **trigrams**. Looking at word sequences rather than isolated tokens makes it possible to capture context and nuance cues that are lost at the unigram level. This approach is particularly relevant in political reporting, where framing is often practiced through modifiers and titles rather than through overtly emotional vocabulary.

For instance (exaggerated example), references to Nicolás Maduro can illustrate this:

- “*Maduro*” on its own is largely neutral and appears across outlets.
- “*Dictator Maduro*” introduces a strong evaluative frame.
- “*President Maduro*” reflects a more institutional or diplomatic framing.

By extracting and comparing frequent bigrams and trigrams across ideological groups, this analysis aims at identify recurring framing patterns. This lexical perspective provides an initial idea of of how the same event is narrated differently. Later in this project, we will apply more complex sentiment / topic-based models.

The Metric: Fixing the Data Imbalance

Our dataset has more Left-wing articles than Right-wing articles. If we simply counted words, the Left would dominate the results. To fix this, we use **Document Frequency (DF)**:

1. We calculate the **Percentage of articles** in which a phrase appears.
2. We look at $DF_{Right} - DF_{Left}$
3. For example, if a phrase appears in 60% of Right articles and 10% of Left articles, the score is **+50** (Strongly Right).

Subtracting the Noise

Initial frequency analyses showed that a large share of high-frequency tokens was not informative. UI-related noise and irrelevant entities were dominating the results and obscuring framing-related patterns.

To address this, we applied an **aggressive stopwords filter** that removed:

1. **Scraping noise**, such as *“twitter”*, *“pic”*, *“video”*, or *“subscribe”*.
2. **Common entities**, including *“United States”*, *“New York”*, and *“Washington”*.
3. **Core subject entities**, namely *“Maduro”* and *“Venezuela”*, since all articles are already about them.

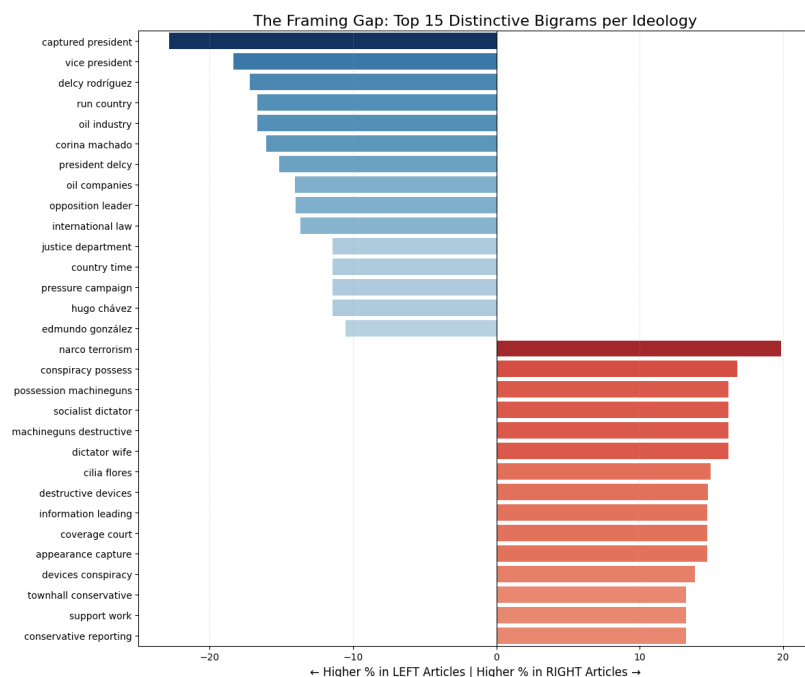
This filtering step allows the analysis to focus on how the event is framed, rather than on repeated or structurally irrelevant terms.

The Visualization Strategy

To make framing differences immediately visible, we rely on two complementary visualizations.

1. First, we use **different word clouds** to highlight vocabulary that is distinctive to each ideological group.
The blue cloud represents phrases that appear proportionally more often in left-wing articles, while the red cloud highlights phrases more common in right-wing coverage. This side-by-side view provides an intuitive overview of how language diverges across ideologies without relying on numerical thresholds.
2. Second, we use a **divergence bar plot** to provide a more precise comparison. The plot displays the top 15 most distinctive bigrams for each ideology, ranked by the difference in document frequency percentages. Phrases on the left side of the axis are more characteristic of left-wing outlets, while phrases on the right are more characteristic of right-wing outlets.

Together, these visualizations allow us to observe both broad lexical patterns and the most salient framing differences in a compact and interpretable way.



Interpretation

We successfully removed the noise. By normalizing frequencies using **article percentage**, we eliminate the bias introduced by the unequal size of the two groups. This ensures that the observed differences reflect framing choices rather than dataset volume.

How to read the bar plot:

The X-axis represents the difference in percentage points between right-wing and left-wing articles.

For example, if “*narco terrorism*” has a value of +10, this means that the phrase appears in 10% more right-wing articles than left-wing articles.

Analysis of the word cloud and the bar plot:

- **Right-Wing (Red):**
Right-leaning outlets place strong emphasis on criminality and law enforcement. Terms such as “*conspiracy*,” “*narco terrorism*,” and “*machineguns*” appear more frequently, framing the capture as a successful crackdown on a dangerous criminal network.
- **Left-Wing (Blue):**
Left-leaning outlets focus more on institutional and procedural aspects. Phrases related to “*international law*,” “*opposition leader*,” or legal authority are more prevalent, framing the event as a complex geopolitical action with diplomatic and legal implications.

Going further: Count of Unique Words

Another interesting thing to look at is the unique vocabulary usage in Left/Right-leaning articles to determine whether the narrative they carry relies on complex explanations or repetitive messaging.

Because vocabulary growth is non-linear (Heaps' Law, according to the internet), we cannot simply divide by total word count to account for the class imbalance (we have more left-leaning articles than right-leaning).

The simplest method consists in randomly sampling as many article from the Left as we have from the Right.

ideology	Unique Words (Adjusted)
0 Left	8010
1 Right	7218

Interpretation

Running this manually several times (to account for random sampling), we observe that the unique count for the Left fluctuates around 8000, whereas it is 7200 for the Right.

So, the Left has a moderately larger lexicon (~1.14x).

- This suggests a **broader scope of topics** or more **varied terminology**.

- In contrast, the Right's slightly more concentrated vocabulary implies a **higher repetition of specific terms or a narrower thematic focus**. The difference is statistically present but too small to definitively claim a disparity in messaging complexity.

Part 3: Sentiment Analysis (Baseline vs. Transformer)

We want to look at sentimental/affective framing to **determine whether ideological divergence manifests as a difference in emotional tone** or if, on the contrary, it is only a matter of distinct surface-level vocabulary.

To do this, we will use 2 approaches that each have their advantage and compare their results.

Methodology: The "Lead Paragraph" Hypothesis

In journalism, the most emotionally charged and framing-heavy information is typically concentrated in the **lede**, meaning the opening paragraphs of an article. This is where journalists set up the tone, establish responsibility, and define how the reader should interpret the event.

1. **Baseline (VADER):**

VADER is a lexicon-based sentiment model that computes sentiment by aggregating the predefined valence scores of individual words. It is fast and well suited for short texts, but it struggles with complex context, irony, or politically loaded language where meaning depends on phrasing rather than isolated terms.

2. **Transformer (RoBERTa):**

RoBERTa is a transformer-based model that is fine-tuned for sentiment analysis. Unlike lexicon-based approaches, it captures contextual meaning and handles negation or subtle phrasing (e.g., *"the plan was not bad"*). This makes it more appropriate for political news, where sentiment is often implicit.

Constraint:

BERT-based models are limited to approximately 512 tokens per input.

Solution:

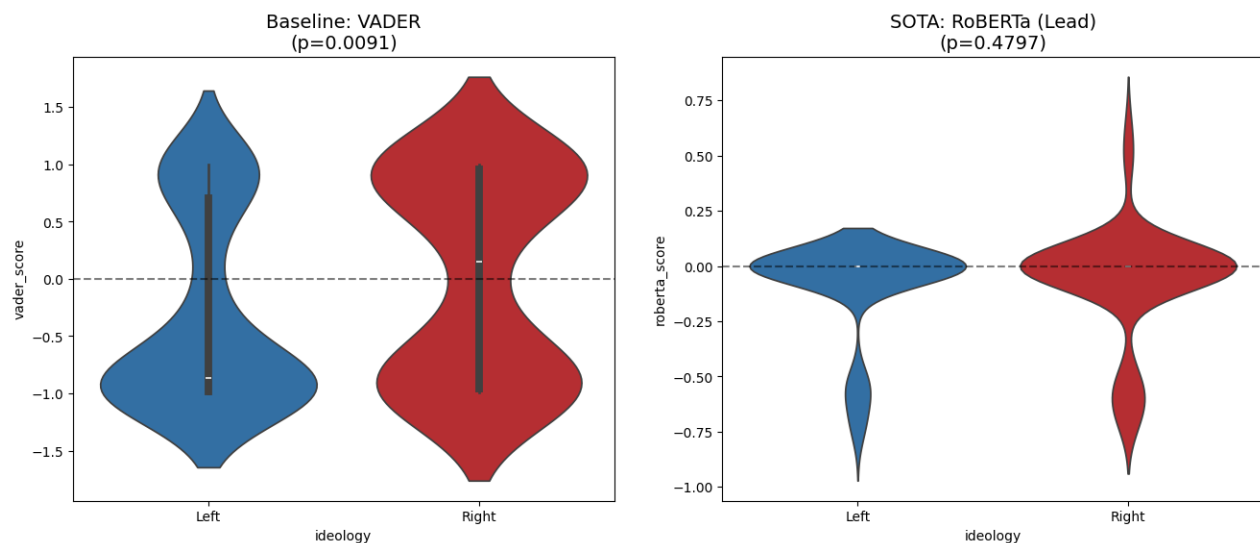
Rather than averaging sentiment across the entire article, we restrict the analysis to the first 512 tokens. Averaging sentiment over long articles tends to push scores toward neutrality due to central tendency. Focusing on the lede preserves the framing signal, as this section contains the narrative "hook" that shapes the reader's interpretation from the outset.

Statistical Testing

We want to know if the two unpaired (independent) samples (left/right articles) are drawn from the same (sentiment) distribution. By comparing their rank sums (checking for stochastic dominance) rather than their raw means, we cannot simply compare the means of the sentiment distributions because sentiment data is rarely normally distributed.

We will thus use the **Mann-Whitney U Test** which is the non-parametric equivalent to the t-test. It is also adapted because it is robust to class imbalance (which is our case).

Based on the p-value obtained, we will determine if the distributions of the Left and Right scores are statistically different.



Understanding our plots

1. Significance (P-Value):

- If $p < 0.05$: The difference in sentiment between Left and Right is **real**, not random chance.
- If $p > 0.05$: We cannot claim a difference in tone.

2. Distribution Shape (The Violin):

- **Fat middle**: Most articles are neutral.

- **Fat bottom:** Heavy negative framing (fear/anger).
- **Fat top:** Positive framing (celebration/support).
- **The Median** of the distribution is indicated by a white dot.

Interpretation

VADER Analysis (Lexicon-Based)

1. Statistical signal:

The p-value (0.0091) indicates a statistically significant difference between left-wing and right-wing sentiment distributions. We **reject** the Null hypothesis that the samples draw from the same distribution.

2. Distribution shape:

Both groups show bimodal distributions, with distinct positive and negative clusters, a pattern typical of VADER's lexicon-based scoring.

The left-wing distribution (blue) is bottom-heavy, reflecting frequent use of negative vocabulary such as *"crisis"* or *"concern"*.

The right-wing distribution (red) is more balanced, with a stronger positive cluster driven by terms like *"justice"* and *"victory"*.

3. Median shift:

The medians (white markers) highlight this divergence.

The left-wing median lies in the negative range, while the right-wing median crosses into positive territory, indicating more celebratory language on the right.

RoBERTa Analysis (Context-Aware)

• Statistical signal:

The high p-value (0.4797) indicates no statistically significant difference between left-wing and right-wing sentiment distributions. We **fail to reject** the Null hypothesis that the samples draw from the same distribution.

• Distribution shape:

Both distributions are unimodal and tightly compressed around the neutral axis (0.0), forming inverted "T"-like shapes.

The downward-extending tails indicate that negative sentiment appears as an outlier rather than a dominant pattern.

• Median alignment:

The medians (white markers) are identical and centered at zero.

This shows that once contextual meaning is accounted for, the apparent emotional gap

disappears, and both sides adopt a largely neutral journalistic tone in their opening paragraphs.

Synthesis: The contrast between the two sentiment plots highlights the difference between **lexical bias** and **tonal reality**. While left-wing and right-wing outlets rely on different vocabularies—triggering divergent results in the lexicon-based VADER analysis—they ultimately write with a similar level of professional detachment, as captured by the contextual RoBERTa model.

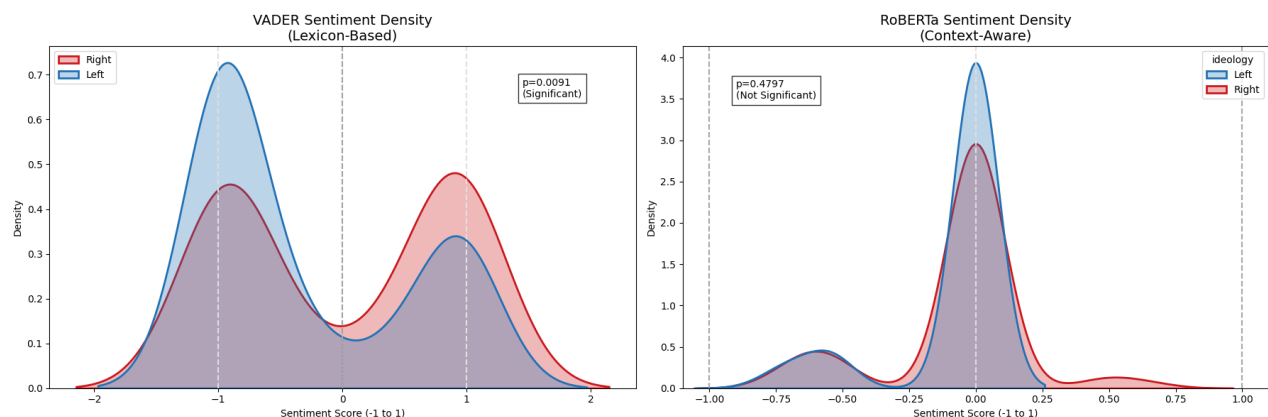
We found this discrepancy particularly interesting. To further visualize it, we plot **normalized distribution curves** using Kernel Density Estimates (KDEs). These plots allow the left-wing and right-wing sentiment distributions to be overlaid directly, making it easier to observe where they overlap and where subtle divergences remain.

Comparative Distribution Analysis (VADER vs. RoBERTa)

To better understand why VADER detects a sentiment difference while RoBERTa does not, we directly overlay the sentiment distributions for left-wing and right-wing articles.

We use **normalized Kernel Density Estimates (KDEs)** to compare the *shape* of the sentiment distributions rather than raw counts. This allows both ideologies to be visualized on the same scale, making differences and overlaps immediately visible. Areas where the curves overlap indicate agreement between ideological groups, while separation highlights divergence.

The contrast between the two plots reinforces our earlier findings. VADER produces clearly separated distributions, reflecting differences in vocabulary usage. In contrast, the RoBERTa distributions almost entirely overlap around the neutral range, confirming that once context is taken into account, both sides exhibit a similar, neutral journalistic tone.



Interpretation

As we found in the violin plots,

1. **VADER (Left plot):**

The blue curve (Left) is clearly shifted toward the negative side compared to the red curve. This aligns with the statistical result and confirms that right-wing outlets rely more on positively charged vocabulary (e.g., “victory,” “justice”), while left-wing outlets favor more cautious or negative terms.

2. **RoBERTa (Right plot):**

The two curves overlap almost entirely. While the red curve (Right) shows a slightly longer positive tail, the overall distributions remain centered around neutrality. This indicates that despite lexical differences, both sides maintain a similar, standard “serious news” tone in their reporting.

Conclusion

This comparison highlights a key methodological difference between lexicon-based and transformer-based sentiment models. Lexicon models such as VADER are highly sensitive to individual keywords and tend to merge **institutional or procedural language** with negative sentiment. In contrast, transformer models like RoBERTa account for context and **infer sentiment** from the overall evaluative stance of the text.

Consider the following example:

“Venezuelan authorities confirmed the capture of Nicolás Maduro following negotiations with military leadership. He was transferred to a secure facility pending legal proceedings.”

VADER interprets words such as “capture”, “authorities”, and “legal proceedings” as **negative or punitive**, resulting in an overall negative sentiment score. RoBERTa, however, detects no explicit emotional or evaluative language—no adjectives indicating approval, condemnation, or affective stance—and therefore assigns a **neutral score**.

This illustrates why VADER captures **lexical bias**, while RoBERTa better reflects **tonal reality**. The two models are not contradictory, but instead reveal different layers of media framing: one driven by word choice, the other by narrative intent.

Part 4: Topic Modeling (The "Framing" Analysis)

Beyond observing *how* left-wing and right-wing outlets frame the event, we also aim to understand *why* these differences emerge. To address this, we turn to **topic modeling**, which allows us to identify the underlying themes structuring the coverage.

We choose **Non-Negative Matrix Factorization (NMF)** rather than Latent Dirichlet Allocation (LDA). LDA assumes that documents are probabilistic mixtures of multiple topics, which can blur distinctions in relatively focused texts such as news articles. In contrast, NMF tends to produce more distinct and interpretable topics, making it better suited for identifying specific frames.

This approach is particularly useful for our objective, as it helps separate clearly defined narratives, such as framing the event in terms of *human rights and international law* versus *criminal activity and drug trafficking*.

Methodology: NMF (Non-Negative Matrix Factorization)

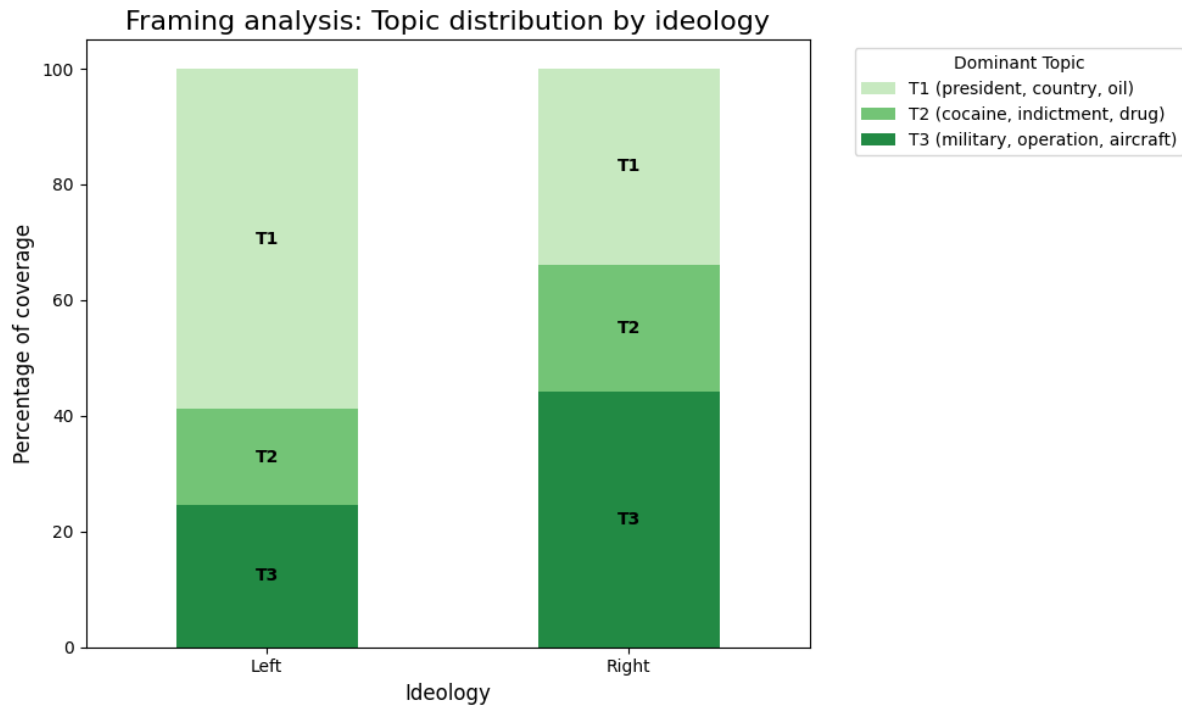
1. Preprocessing

Before fitting the model, the text is filtered to retain only **nouns and adjectives**. This step removes generic action verbs such as “*said*” or “*went*”, which are frequent but uninformative, and shifts the focus toward actors, objects, and descriptive language that define framing.

2. Metric

For each article, we identify its dominant topic and compute a **topic share** by ideology. This metric represents the percentage of left-wing or right-wing coverage devoted to each topic, allowing for a direct comparison of thematic emphasis between the two groups.

```
-- Discovered Topics --  
Topic 1: president, country, oil, international, administration  
Topic 2: cocaine, indictment, drug, conspiracy, narco  
Topic 3: military, operation, aircraft, explosions, helicopters
```



Interpretation

This result confirms the framing hypothesis.

1. Left-Wing framing:

Left-wing coverage is dominated by **Topic 1** (*“President, Country, Oil”*), which accounts for approximately 60% of their articles. This indicates a focus on geopolitics, sovereignty, and institutional legitimacy. The event is framed primarily as a political and diplomatic issue, with attention given to implications for Venezuela, international relations, and energy markets, rather than to the operational details of the capture.

2. Right-Wing framing:

Right-wing coverage is more strongly driven by **Topic 3** (*“Military, Operation”*), which represents around 40% of their articles. **Topic 2** (*“Cocaine, Indictment”*) is also more prevalent than in left-wing coverage. Together, these topics suggest a framing centered on enforcement and security. The event is presented as a successful military and law enforcement operation, emphasizing action (raids, aircraft, arrests) and justification (drug trafficking), while downplaying the presidential or diplomatic context highlighted in Topic 1.

Conclusion:

In broad terms, left-wing outlets frame the event as a political crisis, whereas right-wing outlets frame it as a drug bust.

Convergence or Polarization?

We looked at the highest weight in the W matrix for each article, to interpret the intensity of the framing. We calculated the spread from our Topic table, to interpret the Polarization Gap.

```
-- Framing Confidence --
ideology
Left    0.189050
Right   0.187617
Name: framing_confidence, dtype: float64

-- Polarization Metrics --
Left's Monopoly on T1 (Procedural): +24.9%
Right's Monopoly on T3 (Militarist): +19.6%
```

1. Discipline (Convergence)

The nearly identical scores indicate that both sides are equally consistent/rigid on their narrative. Neither ideology is more open-minded or scattered than the other.

2. Symmetrical Polarization

They are mirror images. The Left "owns" the Procedural Frame (T1) by a margin of 25%, while the Right "owns" the Militarist Frame (T3) by 20%. They are talking of their preferred topic with similar intensity. This symmetrical polarization suggests a shared reality but mutually exclusive interpretations.

Example of Document Vector

```
-- Example of Document Vector (First Article) --
              TF-IDF Score
sources      0.187530
team         0.174509
operation    0.168908
strike       0.167202
president    0.154833
weather      0.148277
ahead        0.140722
familiar plans 0.140722
timeline     0.140722
national security 0.129119
done
```

This is an example of how an article appears to the model after applying TF-IDF. Instead of raw text, the document is now a **vector of weighted scores**. Note that for report purposes we filtered out the zeros in this set, but a vector is typically very sparse.

A high score means the word is statistically significant for this specific article compared to the rest of the dataset (eg, "*cocaine*" has a high weight, while common words have zero).

SPSS table of Document-Topic Matrix

```
-- Document-Topic matrix --
```

	topic_1	topic_2	topic_3	dominant_topic	ideology
...					
1	0.157185	0.000000	0.064132	1	Left
2	0.190127	0.000000	0.000000	1	Left
3	0.127553	0.000000	0.000000	1	Left
4	0.204373	0.025115	0.211484	3	Left

As suggested in the feedback, we show what the Topic Weights Matrix looks like. Here, each row is a document, and each column is a topic weight, enabling classical statistical analysis.

Part 5: Named Entity Recognition (NER)

After identifying *how* the event is framed lexically and *why* certain themes dominate, we now turn to the question of *who* is emphasized in each narrative. Named Entity Recognition allows us to move from abstract framing to concrete actors.

- Lexical analysis tells us **which words are used**.
- Sentiment analysis tells us **how the event is described**.
- NER helps identify **who is credited, blamed, or positioned at the center of the story**.

This section identifies the main “**protagonists**” highlighted by left- and right-wing outlets by comparing the entities each side emphasizes. Using **spaCy’s Named Entity Recognition**, we analyze which actors (people, organizations, locations) are foregrounded to understand how responsibility and agency are distributed across narratives.

Methodology: The "Protagonist" Check

- **Model:** We use spaCy’s Named Entity Recognition to detect and classify proper nouns, focusing on **PERSON**, **ORG**, and **GPE** (geo-political entities such as countries or states).
- **Hypothesis**
 - Right-wing media tends to personalize the event by highlighting individual actors (e.g. Flores, Rubio).
 - Left-wing media emphasizes institutions, framing the events through organizations such as UN, ICC or DOJ.
- **Preprocessing constraints**

spaCy relies on raw, case-sensitive text with punctuation. As a result, we do not lowercase or strip punctuation before applying NER, since this would degrade entity detection (e.g. “*White House*” as an organization vs “*white house*” as a building).
- **Adjustments and corrections**
 - During our initial tries, Spacy was making mistakes (eg, labeling Trump or Getty Images as organizations). We addressed this by applying ban lists and manual remapping for known cases and *people_as_orgs_fix* (force mapping).
 - During our initial tries, Spacy was counting similar entities as distinct (eg, *Trump* vs *Donald Trump*). Also, SpaCy handles accents but does not merge things like: *Nicolas*, *Nicolás*.

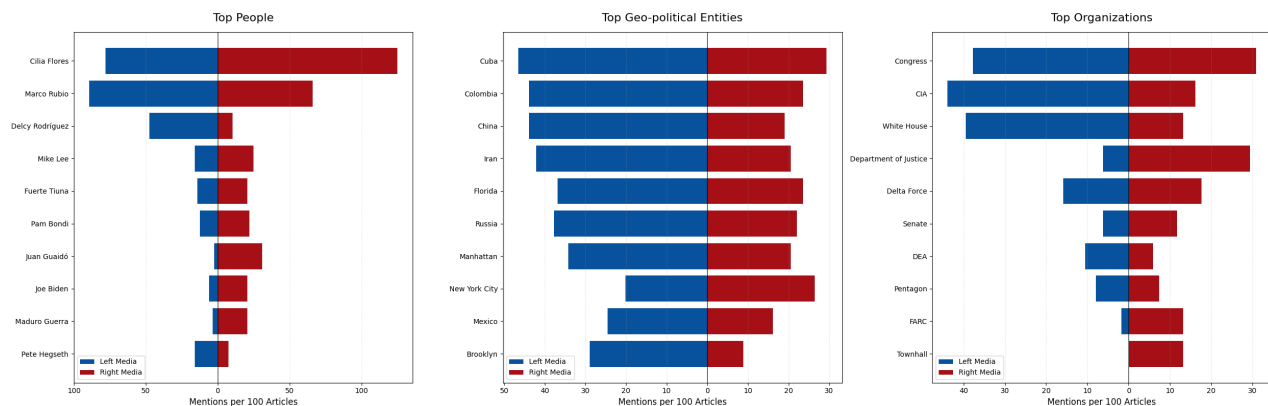
We could not reuse `fft` (from part1) because this only fix encoding errors (e.g. merging *NicolÃs* with *Nicolás*). We thus implemented an *entity_map* to fix it. Also, we removed high-frequency

terms like "*Venezuela*", "*US*", "*Trump*", "*Maduro*" because they appear in 100% of articles and as such do not bring any interesting value.

Visualization choice

We use a **Population Pyramid** visualization (often used in demographics).

- **Metric:** Mentions per 100 Articles (Normalized). This is crucial because we have more Left-wing articles; raw counts would be misleading.
- **Center:** Entities appear equally on both sides (Consensus).
- **Edges:** Entities skewed heavily to one side (Partisan focus).



Interpretation

1. Persons

Left-wing outlets emphasize **Marco Rubio**, framing him as a policy architect, while **right-wing** outlets focus on **Cilia Flores**, framing the regime as a criminal network rather than a government. Main figures (Trump, Maduro) were excluded.

2. Geo-political Entities

The presence of **Manhattan** and **Brooklyn** alongside global actors such as **Cuba**, **China**, and **Iran** reflects a dual framing of the event as both an international geopolitical issue and a New York–based legal case. Core locations (US, Venezuela, New York, Caracas) were excluded.

3. Organizations

- **Right Skew: DOJ, Delta Force, FARC.** The Right frames this as a military/law enforcement operation against narco-terrorism.
- **Left Skew: CIA, White House, Congress.** The Left frames this through the lens of intelligence failures and legislative oversight.

Honorable Mentions

Here are things we tried but did not keep in our final code/report for the sake of (relative) brevity.

1. Baseline: TextBlob vs. VADER

We initially tested the TextBlob model but found its polarity scoring too fade for political nuance. We switched to VADER because it better isolates the lexical nuances of specific vocabulary.

2. Context: Longformer vs. RoBERTa

We tried to score full articles using Longformer (can look at 4096 tokens), but the framing signal was diluted. This is supported by "Inverted Pyramid" principle of journalism, which dictates that the framing is always concentrated in the Lede (introduction), and the rest is more neutral. We ended up sticking to RoBERTa which focuses on the first 512 tokens.

Limitations

- Only English/US Sources: We miss local Venezuelan perspectives by limiting the scope to US media.
- Small & Unbalanced Data: The dataset is relatively small (182 articles) because we chose quite a specific event. We tuned the scraping method/instructions but could probably have improved it further.
- Wordcloud: Simple word frequency can be misleading because they don't see negatives (eg counting "good" inside "not good"). We used word pairs (Bigrams) to fix this, but it doesn't take into account long-range dependencies, so we have to remain careful in our interpretation. This is why the BERT analysis is more interpretable.
- Truncating texts: We only analyzed the first ~512 words. As stated in Honorable Mentions, this is because we saw that using the full articles returned rather inconclusive results. However, we might miss later tone changes in some articles.

Conclusion

RQ: “How do left-wing and right-wing English-language news media differ in sentiment and topic framing when reporting on US actions against Nicolás Maduro?”

Our multi-approach analysis reveals that while the **tone** of reporting remains professional across the spectrum (RoBERTa found no emotional difference), the **reality** constructed by the media is fundamentally split.

By selecting specific topics (NMF) and entities (NER):

- The **Right-Wing** media constructs a narrative of a **Tactical Victory** against a criminal enterprise.
- The **Left-Wing** media constructs a narrative of a **Strategic Geopolitical Crisis** with legal complexities.

GenAI Usage & Other Sources

We mostly use GenAI for short and straightforward indications like “how do we go about fixing x issue?”. Then, we referred to the documentation of the libraries/functions, or found info on how to use this or that arguments.

Getting help on plots documentation

- I want to compare visually the distribution of sentiment scores between two groups of unequal sizes. What is the best plot?

You should use a **Violin Plot** (`sns.violinplot`). It combines a box plot with a kernel density estimate, allowing you to see the distribution shape (modes, skewness) and probability density for each group side-by-side, which is better than a simple boxplot for detecting bimodality (peaks).

- How to normalize 2 KDE plots so that the total area is same for each? I have 2 distributions but not of the same size.

You must use the parameter `common_norm=False` in `sns.kdeplot`. This ensures each distribution is normalized independently (area=1 per group), allowing for a direct comparison of the *shape* of sentiment rather than raw frequency, effectively correcting for your class imbalance.

- How to create a horizontal bar chart in Matplotlib where one category goes left from the center and the other goes right?

To create a "Butterfly Chart" (Population Pyramid), plot the Left-wing values as **negative numbers** (`-df['Left']`) so they extend leftwards on the X-axis. Then, use `ax.set_xticklabels([abs(x) ...])` to hide the negative signs so the labels appear positive on both sides.

- Check in the Matplotlib documentation how to add labels to the stacked rectangles of a stacked bar chart.

In Matplotlib, you can iterate through `ax.containers`. Since a stacked bar chart builds bottom-up, the containers match your topic order; loop through them and use `ax.bar_label(container, label_type='center')` to place the percentage values inside the segments.

Statistical testing

- How do I know if I need to use a non-parametric statistical test? My case is I have 2 sample of numerical data (sentiment scores) of different sizes, and I want to test whether they come from the same distribution.

Since sentiment scores are rarely normally distributed and your sample sizes are unequal/unpaired, you cannot use a standard T-test. You should use the **Mann-Whitney U Test** (`scipy.stats.mannwhitneyu`), which compares the *rank sums* to determine if one distribution is stochastically greater than the other without assuming normality.

Specific Tools documentation

- Explain to me how to use Spacy for Topic Modeling. I want to find the main topics in left-Leaning vs Right-leaning articles, what should my code workflow be? And what about for entity detection (GPE/person/Institutions)?

For Topic Modeling, use Spacy only to **filter for Nouns and Adjectives** (POS tagging) to sharpen the semantic content, then pass that filtered text to Scikit-Learn's **NMF** model (TF-IDF vectorizer). For Entity Detection, use Spacy's `nlp.pipe(docs)` for efficiency, and access `doc.ents` to extract specific labels like `PERSON`, `ORG`, or `GPE`.

- What is the parser arg in Spacy? In doing NER and I don't understand why it's taking so long.

The parser is the component responsible for analyzing grammatical structure and dependencies (like linking a subject to its verb), which is computationally heavy. Since you are strictly performing Named Entity Recognition (NER) and do not need syntactic trees, you should disable it by calling `nlp.pipe(docs, disable=['parser'])` to significantly speed up your processing pipeline.

- How to use Regex to remove artifacts like copyright tags etc in scraped text data?

Create a list of artifact strings (e.g., "Copyright", "Follow us"), join them with a pipe `|` to create a single "OR" pattern, and use `re.sub(pattern, '', text, flags=re.IGNORECASE)`. This is significantly faster and cleaner than running a loop of individual replacements.

Main documentation sources used for further info & troubleshooting:

- Visualization (Matplotlib & Seaborn)

<https://seaborn.pydata.org/generated/seaborn.violinplot.html>

<https://seaborn.pydata.org/generated/seaborn.kdeplot.html>

https://matplotlib.org/stable/api/as_gen/matplotlib.axes.Axes.bar_label.html

https://matplotlib.org/stable/gallery/lines_bars_and_markers/barh.html

- WordCloud: used `recolor()` with a custom function to make the clouds specifically Blue/Red for consistency

http://amueller.github.io/word_cloud/generated/wordcloud.WordCloud.html#wordcloud.WordCloud.recolor

- NLP & Machine Learning (Spacy Scikit-learn)

<https://spacy.io/usage/linguistic-features#named-entities>

<https://spacy.io/usage/processing-pipelines#disabling> # parser arg to speed up

<https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.NMF.html> # nmf

[https://scikit-](https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html)

[learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html](https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html)

#Tf-idf for n-grams

https://huggingface.co/docs/transformers/main_classes/pipelines # for huggingface pipe for roberta

<https://www.nltk.org/api/nltk.sentiment.vader.html> # vader polarity scores for baseline

<https://docs.python.org/3/library/re.html> # regex to clean

<https://ftfy.readthedocs.io/en/latest/> # ftfy to fix encoding errors